

full study:
2 assets ×
10 seeds

per family × asset × seed:
15 walk-forward folds, processed independently

random search (primary engine)
100 candidates per fold
seed: SeedSequence(base_seed,
spawn_key = blake2b(stream label))

genetic algorithm (cross-check)
identical evaluation budget
budget-parity ledger records
evaluations and wall-time per fold

select on validation Sharpe
(train fit → validation ranking;
test window never touched)

select on validation Sharpe
same folds, same costs,
same seeds discipline

winner re-run once on the fold's test window
→ OOS segments concatenated
verdicts read on random search only